

Urinary system disorders:
Urinary tract infection, cystitis, abnormal renal function, acute renal failure

Liver and biliary system disorders:
Hepatic enzymes increased, bilirubinemia, infectious hepatitis, fatty liver

Vision disorders:
Cataract, conjunctivitis

Post-marketing experience
The following events have been reported spontaneously from worldwide postmarketing experience:

Immune System disorders:
Hypersensitivity reactions, anaphylactoid reactions, serum sickness

Central and Peripheral Nervous System disorders:
Confusion, hallucinations, taste disorders

Hepato-biliary disorders:
Hepatitis (non-infectious), acute liver failure

Blood and lymphatic system disorders:
Cases of bleeding with fatal outcome (especially intracranial, gastrointestinal and retroperitoneal hemorrhage), Thrombotic thrombocytopenic purpura (TTP), severe thrombocytopenia, agranulocytosis, granulocytopenia, aplastic anaemia/pancytopenia, Anaemia, conjunctival, ocular and retinal bleeding.

Respiratory, thoracic and mediastinal disorders:
Bronchospasm; interstitial pneumonitis

Skin and subcutaneous tissue disorders:
Angioedema, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, lichen planus

Renal and urinary disorders:
Glomerulonephritis, increased creatinine levels

Vascular disorders:
Vasculitis, hypotension

Gastrointestinal disorders:
Colitis (including ulcerative or lymphocytic colitis); pancreatitis; stomatitis

Musculoskeletal, connective tissue and bone disorders:
Myalgia

OVERDOSAGE
No adverse effects have been reported after ingestion of up to 14 tablets of clopidogrel 75 mg; bleeding time after 6 tablets of clopidogrel 75 mg was prolonged by about the same amount as with the usual therapeutic dose of 1 tablet of clopidogrel 75 mg.

The lethal single oral dose in mice and rats was 1500 or 2000 mg/kg and in baboons was 3000 mg/kg. Symptoms of acute toxicity included vomiting (in baboons), prostration, difficult breathing, and gastrointestinal hemorrhage in all species. Overdose with clopidogrel may lead to prolonged bleeding time and subsequent bleeding complications. Platelet transfusion would be appropriate to reverse the undesirable effects of clopidogrel if quick reversal is required.

STORAGE
STORE BELOW 30°C, PROTECT FROM MOISTURE

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

PACKING
Blister Strips of 10 x 10's and 3 x 10's

Date of revision: January 2017

Product Registration Holder/
Manufactured by :
RANBAXY (MALAYSIA) SDN. BHD.
Lot 23, Bakar Arang Industrial Estate,
08000 Sungai Petani, Kedah,
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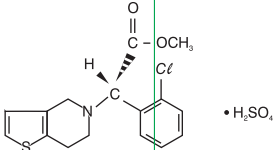
COMPOSITION

CERUVIN TABLETS
Each film-coated tablet contains:
Clopidogrel bisulphate
equivalent to Clopidogrel75 mg

PRODUCT DESCRIPTION
Pink coloured, round, film-coated tablets with 'C' debossed on one side and plain on the other side.

DESCRIPTION

CERUVIN TABLETS contain clopidogrel bisulphate. Clopidogrel inhibits platelet aggregation. It is a thienopyridine derivative which is chemically related to ticlopidine. It is chemically designated as methyl (+)-(S)- α -(O-(chlorophenyl)-6, 7-dihydrothieno [3, 2-c] pyridine-5(4H)-acetate sulphate (1:1). The empirical formula of clopidogrel bisulphate is $C_{16}H_{14}ClNO_2S \cdot H_2SO_4$ and its molecular weight is 419.9. The structural formula of clopidogrel bisulphate is given below



CLOPIDOGREL BISULPHATE

PHARMACOLOGY

• **Mechanism of Action**
Clopidogrel is a thienopyridine derivative which inhibits platelet aggregation. In people with established atherosclerosis, the incidence of morbid events (such as unstable angina, myocardial infarction, stroke, transient ischemic attacks, or the need for vascular bypass or angioplasty) has been shown to decrease with the use of drugs inhibiting platelet function. This demonstrates the involvement of platelets in the initiation and/or evolution of such events, and that inhibition of platelet aggregation can reduce the morbidity rate.

• **Pharmacokinetics**
Absorption: The absorption of clopidogrel is rapid but incomplete after oral administration; absorption appears to be at least 50%. Intake of food does not significantly modify the bioavailability of clopidogrel.

The plasma levels of the parent compound are low and fall below the quantification limit beyond 2 hours after dosing, even after repeated 75 mg doses of clopidogrel.

Metabolism: Clopidogrel is a prodrug which is extensively metabolized by liver. The main circulating metabolite is a carboxylic acid derivative (inactive) which accounts for approximately 85% of the circulating drug-related compounds in plasma. Peak plasma concentration of the carboxylic acid derivative is approximately 3 mg per liter, achieved about 1 hour after dosing (on repeated doses of 75 mg). This derivative is the active metabolite but has not been identified in plasma.

The oxidative step is regulated primarily by cytochrome P450 isoenzymes 2B6, 3A4, 1A1, 1A2 and 2C19.

The glucuronide derivative of the carboxylic acid metabolite is also observed in plasma and urine.

The pharmacokinetics of the main circulating metabolite are linear in the dose range of 50 to 150 mg.

Clopidogrel and its carboxylic acid derivative have very high protein binding (98% and 94%, respectively). The binding is nonsaturable in vitro up to a concentration of 100 µg/mL.

Elimination: Both urine and faeces are involved in the excretion of clopidogrel. After both single and repeated administration, the elimination half-life of the main circulating metabolite has been seen to be 8 hours. 2% of radiolabelled drug binds to platelets by covalent bond, which accounted for a half-life of 11 days.

• **Pharmacogenetics**
Clopidogrel is activated by several polymorphic CYP450 enzymes. CYP2C19 is involved in the formation of both the active metabolite and the 2-oxo-clopidogrel intermediate metabolite. Pharmacokinetics and antiplatelet effects of clopidogrel's active metabolite differ according to CYP2C19 genotype. The CYP2C19*1 allele corresponds to fully functional metabolism while the CYP2C19*2 and CYP2C19*3 allele account for 85% of the reduced function alleles in whites and 99% in Asians. Other alleles associated with reduced metabolism include CYP2C19*4, *5, *6, *7 and *8, but these are less frequent in general population.

Table- CYP2C19 Phenotype and Genotype Frequency

	Frequency (%)*		
	White	Black	Chinese
Extensive metabolism: CYP2C19*1/*1	74	66	38
Intermediate metabolism: CYP2C19*1/*2 or *1/*3	26	29	50
Poor metabolism: CYP2C19*2/*2 or *2/*3 or *3/*3	2	4	14

*Xie et al. Annu Rev Pharmacol Toxicol 2001; 41: 815-50

Reduced CYP2C19 metabolism in intermediate and poor metabolizers decreases the C_{max} and AUC of active metabolite by 30-50% following 300 or 600 mg loading dose and 75 mg maintenance dose. Lower active metabolite exposure results in less platelet inhibition or higher residual platelet activity. Diminished antiplatelet responses to clopidogrel have been described for intermediate and poor metabolizers in various clinical studies. The relative difference in antiplatelet response between genotype groups varies across studies depending on the method used to evaluate response, but is typically greater than 30%.

The association between CYP2C19 genotype and clopidogrel treatment outcome has been reported. In some of the reported studies, cardiovascular event rates did not differ significantly by genotype. But in other reported studies, patients with an impaired metabolizer status (immediate and poor combined) had a high rate of cardiovascular events (death, myocardial infarction and stroke) or stent thrombosis compared to extensive metabolizers. In another study, the increased event rate was observed only in poor metabolizers. Pharmacogenetic testing can identify genotypes associated with variability in CYP2C19 activity. There may be genetic variants of other CYP450 enzymes with effects on the ability to form clopidogrel's active metabolite. • Special Populations Geriatric: As compared to young healthy volunteers, the plasma concentrations of the carboxylic acid derivative are significantly higher in elderly (≥75 years). However, the higher plasma levels have not led to differences in platelet aggregation and bleeding time. Dosage adjustment is not required in the elderly patients. Renal Insufficiency: After repeated doses of 75 mg/day, plasma levels of carboxylic acid derivative have been found to be lower in patients with severe renal impairment (Creatinine clearance (CCr) 5 - 15 mL/min) than in subjects with moderate renal impairment (CCr 30 - 60 mL/min) or healthy subjects. Although inhibition of ADP-induced platelet aggregation have been found to be lower (25%) compared to that reported in healthy subjects, the prolongation of bleeding time have been found to be similar compared to that of healthy subjects receiving 75 mg/day clopidogrel. Hepatic Insufficiency: After repeated dose of 75 mg/day for 10 days in patients with severe hepatic impairment, inhibition of ADP induced platelet aggregation was similar to that observed in healthy subjects. The mean bleeding prolongation was also similar in two groups. Gender: The plasma concentration of the carboxylic acid derivative (main circulating metabolite) does not show significant difference between males and females. The incidence of clinical effects, undesirable events and abnormal laboratory values has been found to be comparable in both men and women. Race: The prevalence of CYP2C19 alleles that result in intermediate and poor CYP2C19 metabolism differs according to race/ethnicity (See PHARMACOLOGY: Pharmacogenetics). INDICATIONS CERUVIN (clopidogrel 75 mg tablets) is indicated for prevention of atherothrombotic events in patients suffering from: • Myocardial infarction (from a few days until < 35 days). • Ischemic stroke (from 7 days until < 6 months) or established peripheral arterial disease; • Non-ST segment elevation acute coronary syndrome (unstable angina or non-Q-wave myocardial infarction) in combination with Acetylsalicylic Acid (ASA). DOSAGE AND ADMINISTRATION <i>MI, Ischaemic stroke or peripheral artery disease:</i> The recommended daily dose of CERUVIN (clopidogrel 75 mg tablets) is 75 mg as a single daily dose with or without food. <i>Non-ST segment elevation acute coronary syndrome (unstable angina or non-Q-wave myocardial infarction) in combination with Acetylsalicylic Acid (ASA)</i> CERUVIN (clopidogrel 75 mg tablets) treatment should be initiated with a single 300-mg loading dose and then continued at 75 mg once a day (with ASA 75-325 mg daily). Since higher doses of ASA were associated with higher bleeding risk, it is recommended that the dose of ASA should not be > 100 mg. The optimal duration of treatment has not been formally established. Use upto 12 months has been studied and the maximum benefit is seen at 3 months. <i>Dosage in Children and Adolescent:</i> Safety and efficacy in subjects below the age of 18 have not been established. <i>Dosage in Elderly Patients:</i> No dosage adjustment is required for elderly patients. <i>Dosage in Patients with Renal Impairment:</i> No dosage adjustment is required. However, experience is limited in patients with severe renal impairment. CERUVIN (clopidogrel) should be used with caution in this population. <i>Pharmacogenetics</i> CYP2C19 poor metabolizer status is associated with diminished response to clopidogrel. The optimal dose regimen for poor metabolizers has yet to be determined (See PHARMACOLOGY: Pharmacogenetics). PRECAUTIONS • General Antiplatelet drugs including clopidogrel prolongs the bleeding time. Clopidogrel should be used with caution in patients who may be at risk of increased blood loss from trauma, surgery or pathological conditions. If it is desired to eliminate the antiplatelet effects of clopidogrel prior to elective surgery, the drug should be discontinued 7 days prior to surgery. Due to the risk of bleeding and haematological undesirable effects, blood cell count determination and/or other appropriate testing should be promptly considered whenever clinical symptoms suggestive of bleeding arise during the course of treatment. As with other antiplatelet agents, clopidogrel should be used with caution in patients who may be at risk of increased bleeding from trauma, surgery or other pathological conditions and in patients receiving treatment with ASA, nonsteroidal anti-inflammatory drugs, heparin, glycoprotein IIb/IIIa inhibitors or thrombolytics. Patients should be followed carefully for any signs of bleeding including occult bleeding, especially during the first weeks of treatment and/or after invasive cardiac procedures or surgery. The concomitant administration of clopidogrel with warfarin is not recommended since it may increase the intensity of bleedings. In patients having recent TIA or stroke and are at high risk of recurrent ischemic events, the combination of aspirin and clopidogrel has been found to increase risk of blood loss and has not been found to be clinically superior to clopidogrel monotherapy. Clopidogrel prolongs template bleeding time. Caution should be taken in patients having lesions with a propensity to bleed (such as ulcers) and while taken other drugs that might induce such lesions in patients on clopidogrel therapy. There is very limited experience in patients with severe hepatic impairment, who may have bleeding diatheses. Caution should be taken while prescribing/using clopidogrel in these patients. There is very limited experience in patients with severe renal disorder. Caution should be taken while prescribing/using clopidogrel in these patients. Pharmacogenetics: Based on literature data, patients with genetically reduced CYP2C19 function (intermediate or poor metabolizers) have lower systemic exposure to the active metabolite of clopidogrel and diminished antiplatelet responses, and generally exhibit higher cardiovascular event rates following myocardial infarction than do patient with normal CYP2C19 function (See PHARMACOLOGY: Pharmacogenetics). • Pregnancy No evidence of impaired fertility or fetotoxicity has been reported in reproductive studies in rats and rabbits at doses up to 500 and 300 mg/kg/day (respectively, 65 and 78 times the recommended daily human dose on a mg per square meter of body surface area [mg/m²] basis). There are, however, no adequate and well-controlled studies in pregnant women. In view of the lack of data, clopidogrel is not recommended to be used during pregnancy. • Lactation It is not known whether this medicinal product is excreted in human milk, although studies in rats have shown that clopidogrel and/or its metabolites are excreted in the milk. • Pediatric Appropriate studies on the relationship of age to the effects of clopidogrel have not been reported in pediatric population. Safety and effectiveness have not been established • Geriatric Safety and efficacy of clopidogrel has been established in patients 65 years of age and above				DRUG INTERACTIONS Since clopidogrel is metabolized to its active metabolite by CYP2C19, use of drugs that inhibit the activity of this enzyme would be expected to result in reduced drug levels of the active metabolite of clopidogrel and a reduction in clinical efficacy. Concomitant use of drugs that inhibit CYP2C19 (e.g. proton pump inhibitors) should be discouraged. <i>Aspirin and Nonsteroidal Anti-Inflammatory Drugs (NSAIDs):</i> Aspirin has not been found to modify the clopidogrel-mediated inhibition of ADP-induced platelet aggregation, nor has it been found to increase the prolongation of bleeding time induced by clopidogrel. Clopidogrel potentiated the effect of aspirin on collagen-induced platelet aggregation. A pharmacodynamic interaction between clopidogrel and aspirin is possible, leading to increased risk of bleeding. Clopidogrel + aspirin therapy has been administered together for up to one year. <i>Nonsteroidal Anti-Inflammatory Drugs (NSAIDs):</i> Concomitant use of clopidogrel and NSAIDs may increase the risk of gastrointestinal blood loss. Caution should be taken while prescribing NSAIDs in patients on clopidogrel. <i>Heparin:</i> Concomitant use of heparin and clopidogrel has not been found to affect clopidogrel induced inhibition of platelet aggregation. Clopidogrel has not been found to alter the effect of heparin on coagulation, therefore did not necessitate any dose adjustment of heparin. The risk of bleeding can increase due to the possibility of a pharmacodynamic interaction between clopidogrel and heparin. Hence, caution should be taken while using the two drugs concomitantly <i>Warfarin:</i> Due to the increased risk of blood loss, caution should be taken in patient receiving concomitant therapy with warfarin and clopidogrel. <i>Other Concomitant Therapy:</i> Clopidogrel induced inhibition of platelet aggregation has not been found to be significantly influenced by the coadministration of phenobarbital, cimetidine or estrogen. No clinically significant drug interaction have been reported when clopidogrel was coadministered with atenolol, nifedipine, or both atenolol and nifedipine. Clopidogrel has not been found to alter the pharmacokinetics of digoxin or theophylline when coadministered with these drugs. The extent of clopidogrel absorption is not affected by antacids. Clopidogrel may inhibit cytochrome P450 isoenzyme CYP 2C9 and interactions with drug metabolized by this isoenzyme are theoretically possible; it may also inhibit CYP2B6. Therefore clopidogrel may interfere with the metabolism of phenytoin, tamoxifen, tolbutamide, warfarin, torsemide, fluvastatin, and many NSAIDs. Caution should be taken when any of these drugs is coadministered with clopidogrel. No clinically significant undesirable interactions have been reported in patients on clopidogrel therapy who have been concomitantly given a variety of medications including diuretics, β-blockers, ACE inhibitors, calcium antagonists, lipid lowering drugs, coronary vasodilators, antidiabetic agents (including insulin), thrombolytics, heparins (unfractionated and LMWH), GPIIb/IIIa antagonists, antiepileptic agents and hormone replacement therapy. <i>Laboratory Test Interactions</i> None known CARCINOGENICITY/ MUTAGENICITY/ IMPAIRMENT OF FERTILITY There is no evidence to suggest that clopidogrel is tumorigenic, genotoxic, or has any effect on male or female fertility WARNINGS Few cases of thrombotic thrombocytopenic purpura (TTP) have been reported rarely (during post-marketing surveillance) in patients receiving clopidogrel therapy, sometimes after a short exposure (<2 weeks). The condition is characterized by thrombocytopenia, microangiopathic hemolytic anemia, neurological findings, renal dysfunction, and fever. TTP is a serious condition that can be fatal and requires urgent medical care including plasmapheresis. CONTRAINDICATIONS CERUVIN TABLETS (clopidogrel) are contraindicated in the following patients: — Hypersensitivity to clopidogrel. — Active pathological bleeding: conditions such as peptic ulcer or intracranial hemorrhage. — Severe liver impairment — Breast feeding (Refer Lactation section under Precautions) ADVERSE REACTIONS <i>Haemorrhagic disorders:</i> As with other anti-platelet agents, administration of clopidogrel can result in bleeding and haemorrhagic disorders. The most frequent bleeding events are purpura/bruising and epistaxis. Less frequently observed events include haematoma, haematuria, and eye bleeding (mainly conjunctival). The incidence of gastrointestinal haemorrhage and intracranial haemorrhage (experienced with clopidogrel are 2% and 0.4% respectively. <i>Haematological disorders:</i> Neutropenia, leucopenia, agranulocytosis, thrombocytopenia, eosinophilia, anemia aplastic, anemia hypochromic <i>Central and peripheral nervous system disorders:</i> Headache, dizziness, paraesthesia, hypoaesthesia, neuralgia, vertigo, leg cramps <i>Autonomic Nervous System Disorders:</i> Syncope, palpitation <i>Gastrointestinal system disorders</i> Gastric ulcer and duodenal ulcer, dyspepsia, nausea, vomiting, constipation, diarrhoea, abdominal pain <i>Platelet, bleeding and clotting disorders</i> Increase in bleeding time, decrease in platelet count <i>Skin and appendages disorders:</i> Rash, pruritus, purpura, eczema, skin ulceration, urticaria <i>Body as a Whole – general disorders</i> Chest Pain, accidental/inflicted injury, influenza-like symptoms, pain, fatigue, asthenia, fever, hernia, allergic reaction, ischemic necrosis <i>Cardiovascular disorders, general:</i> Edema, hypertension, cardiac failure <i>Heart rate and rhythm disorders:</i> Atrial fibrillation <i>Metabolic & nutritional disorders:</i> Hypercholesterolemia Gout, hyperuricemia, non-protein nitrogen (NPN) increased <i>Musculo-skeletal system disorders:</i> Arthralgia, back pain, arthritis <i>Psychiatric disorders:</i> Depression Anxiety, insomnia <i>Respiratory system disorders:</i> Upper respiratory tract infection; dyspnea; rhinitis bronchitis; coughing, pneumonia, sinusitis, hemothorax			
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